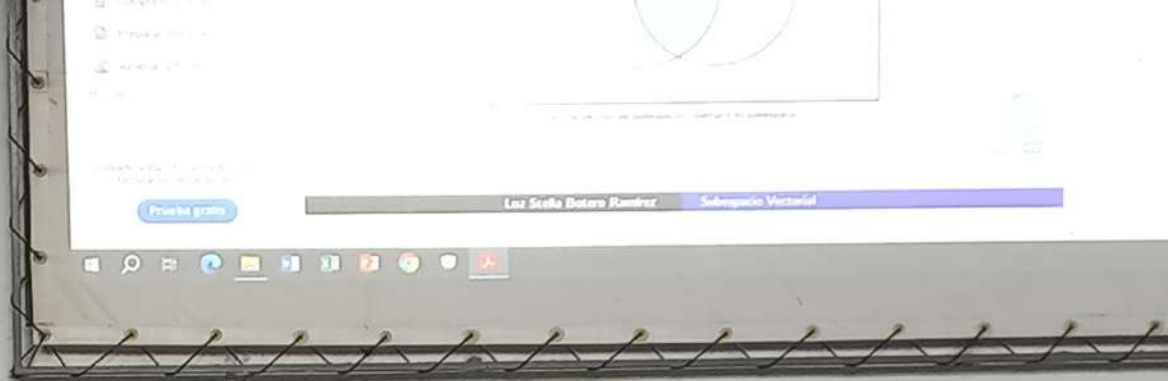




$$H_1 = \left\{ \begin{pmatrix} x \\ y \\ z \end{pmatrix} \in \mathbb{R}^3 / 2x - y - z = 0 \right\}$$

$$H_2 = \left\{ \begin{pmatrix} x \\ y \\ z \end{pmatrix} \in \mathbb{R}^3 / 3x + 2y - z = 0 \right\}$$

$$\textcircled{1} \quad 2x - y - z = 0$$

$$\textcircled{2} \quad 3x + 2y - z = 0$$

$$\begin{pmatrix} 2 & -1 & -1 \\ 3 & 2 & -1 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$A \vec{x} = \vec{y}$

$$\begin{pmatrix} x & y & z & | & 0 \\ 2 & -1 & -1 & | & 0 \\ 3 & 2 & -1 & | & 0 \end{pmatrix} \xrightarrow{2+2} \begin{pmatrix} x & y & z & | & 0 \\ 2 & -1 & -1 & | & 0 \\ 0 & 7 & -3 & | & 0 \end{pmatrix}$$

$$7x - 3z = 0$$

$$x = \frac{3z}{7}$$

$$-y + 2x - z = 0$$
$$y = 2x - z = 2\left(\frac{3}{7}z\right) - z = \frac{6}{7}z - z = -\frac{1}{7}z$$

$y = -\frac{1}{7}z$

vble libre
parámetro

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} \frac{3}{7}z \\ -\frac{1}{7}z \\ z \end{pmatrix} = z \begin{pmatrix} \frac{3}{7} \\ -\frac{1}{7} \\ 1 \end{pmatrix} = \frac{1}{7}z \begin{pmatrix} 3 \\ -1 \\ 7 \end{pmatrix}$$

Escalar el vector por un
escalar de modo que el vector quede
más sencillo

La intersección de los 2 planos:

$$H_1 \cap H_2 = \left\{ \begin{pmatrix} x \\ y \\ z \end{pmatrix} \right\}$$





$$\begin{cases} y-z=0 \\ x+2y-z=0 \end{cases}$$

$$+y \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix}$$

$$\begin{pmatrix} 2 & -1 & 0 \\ -1 & 1 & 0 \end{pmatrix} \xrightarrow{+2+2} \begin{pmatrix} 2 & -1 & 0 \\ 0 & 1 & -2 \end{pmatrix}$$

$$\begin{matrix} -3z=0 \\ 2z \end{matrix}$$

vble libre
parámetro

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} \frac{3}{7}z \\ -\frac{1}{7}z \\ z \end{pmatrix} = z \begin{pmatrix} \frac{3}{7} \\ -\frac{1}{7} \\ 1 \end{pmatrix} = \frac{1}{7}z \begin{pmatrix} 3 \\ -1 \\ 7 \end{pmatrix}$$

Escalar el vector: multiplica por un escalar de forma que el vector quede más sencillo.

La intersección de los 2 planos:

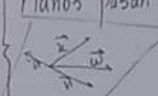
$$H_1 \cap H_2 = \left\{ \begin{pmatrix} x \\ y \\ z \end{pmatrix} \in \mathbb{R}^3 / \frac{x}{3} = -y = \frac{z}{7} \right\} \text{ es subespacio}$$

* H_1, H_2 verificamos si es o no subespacio

Rectas pasan por origen

$$\frac{x}{d_1} = \frac{y}{d_2} = \frac{z}{d_3}$$

Planos pasan por origen



$$A(x-x_0)+B(y-y_0)+C(z-z_0)=0$$

$$ax+by+cz=0$$

$$\begin{matrix} 2x-y-z=0 \\ x+2y-z=0 \end{matrix} \xrightarrow{2x-y-z=0} \begin{matrix} x+2y-z=0 \\ 2x-y-z=0 \end{matrix} \xrightarrow{2x-y-z=0} \begin{matrix} x+2y-z=0 \\ 2x-y-z=0 \end{matrix} \xrightarrow{2x-y-z=0} \begin{matrix} x+2y-z=0 \\ 2x-y-z=0 \end{matrix}$$

Para verificar si la unión es Subespacio.

↪ Tomemos A: punto en H_1 .

$$A: \begin{pmatrix} 1, -1, 3 \\ 0, 3, -3 \end{pmatrix} \in H_1$$

B: pto en H_2

$$B: (1, -1, 1)$$

$$A+B \in H_1 \cup H_2$$

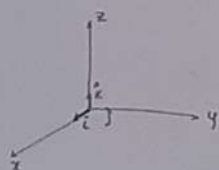
$$(1, -1, 3) + (1, -1, 1) = (2, -2, 4) : C \in H_1 \cup H_2$$

$$C \in H_1 \text{ ó } C \in H_2$$

$$2 \cdot 2 - (-2) - 4 \neq 0 \notin H_1$$

$$3 \cdot 2 + 2(-2) - 4 \neq 0 \notin H_2$$

No es subespacio ya que no cumple la cerradura suma.



$$\begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} = \alpha \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} + \beta \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} + \gamma \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

$$\begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} = 1\hat{i} + 2\hat{j} + 3\hat{k}$$

$$\begin{pmatrix} 4 \\ 0 \\ -3 \end{pmatrix} = \alpha \begin{pmatrix} -2 \\ 6 \\ -6 \end{pmatrix} + \beta \begin{pmatrix} 2 \\ -2 \\ 1 \end{pmatrix} \quad \text{En } \textcircled{2}$$

$$2(3\alpha - \beta) = 0$$

$$3\alpha - \beta = 0$$

$$\begin{pmatrix} 4 \\ 0 \\ -3 \end{pmatrix} = \begin{pmatrix} -2\alpha + 2\beta \\ 6\alpha - 2\beta \\ -6\alpha + \beta \end{pmatrix}$$

$$\begin{aligned} \textcircled{1} \quad & -2\alpha + 2\beta = 4 \\ \textcircled{2} \quad & 6\alpha - 2\beta = 0 \\ \textcircled{3} \quad & -6\alpha + \beta = -3 \end{aligned}$$

$$\text{En } \textcircled{1}$$

$$\begin{aligned} -2\alpha + 2\beta &= 4 \\ -2\alpha + 2(3\alpha) &= 4 \\ -2\alpha + 6\alpha &= 4 \end{aligned}$$

$$4\alpha = 4$$

$$\alpha = 1 \quad \beta = 3$$

$$\begin{pmatrix} 4 \\ 0 \\ -3 \end{pmatrix} = \begin{pmatrix} -2 \\ 6 \\ -6 \end{pmatrix} + 3 \begin{pmatrix} 2 \\ -2 \\ 1 \end{pmatrix}$$

$$H_1 = \left\{ \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} \right\} \in \mathbb{R}^3 / \mathbb{R}$$

$$H_2 = \left\{ \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \right\} \in \mathbb{R}^3 / \mathbb{R}$$

$$H_1 \cup H_2 \text{ es}$$

$$H_1 \cup H_2 = H$$

$$\vec{a} / 3\vec{a}$$

En ②

$$2(3\alpha - \beta) = 0$$

$$3\alpha - \beta = 0$$

$$\boxed{\beta = 3\alpha}$$

En ①

$$-2\alpha + 2\beta = 4$$

$$-2\alpha + 2(3\alpha) = 4$$

$$-2\alpha + 6\alpha = 4$$

$$4\alpha = 4$$

$$\boxed{\alpha = 1} \quad \boxed{\beta = 3}$$

$$\begin{pmatrix} 4 \\ 0 \\ -3 \end{pmatrix} = \begin{pmatrix} -2 \\ 6 \\ -6 \end{pmatrix} + 3 \begin{pmatrix} 2 \\ -2 \\ 1 \end{pmatrix}$$

$$*p(x) = \underset{\substack{\downarrow \\ 1}}{a_0} + \underset{\substack{\downarrow \\ x}}{a_1}x + \underset{\substack{\downarrow \\ x^2}}{a_2}x^2 + \dots + \underset{\substack{\downarrow \\ x^n}}{a_n}x^n$$

$$p(x) = 6 - 3x^2 + 8x^3 \quad \begin{matrix} v_0 = 1 \\ v_1 = x \\ v_2 = x^2 \\ v_3 = x^3 \end{matrix}$$

$$\text{e.l. } p(x) = 6\vec{v}_0 + 0\vec{v}_1 - 3\vec{v}_2 + 8\vec{v}_3$$

$$\begin{aligned} -3 - 6x + x^2 &= \alpha(1 - 2x + x^2) + \beta(3 + x^2) \\ -3 - 6x + x^2 &= (\alpha + 3\beta) + (-2\alpha)x + (\alpha + \beta)x^2 \end{aligned}$$

$$\text{① } \alpha + 3\beta = -3$$

$$\text{② } -2\alpha = -6 \rightarrow \alpha = \frac{-6}{-2} \rightarrow \alpha = 3$$

$$\text{③ } \alpha + \beta = 1$$

$$\text{En ③ } 3 + \beta = 1$$

$$\boxed{\beta = -2}$$

$$\text{En ① } 3 + 3(-2) = 3 - 6 = -3$$

el polinomio es e.l. de los otros 2.

$$-3 - 6x + x^2 = 3(1 - 2x + x^2) - 2(3 + x^2)$$

Ejw. $p(x) = 2 + 3x - x^2$

y sean $p_1(x) = 2 - x + x^2$
 $p_2(x) = 6 - 5x + x^2$

dt si $p(x)$ es c.l. de $p_1(x)$ y $p_2(x)$

$$2 + 3x - x^2 = \alpha(2 - x + x^2) + \beta(6 - 5x + x^2)$$

$$2 + 3x - x^2 = (2\alpha + 6\beta) + (-\alpha - 5\beta)x + (\alpha + \beta)x^2$$

① $2\alpha + 6\beta = 2 \rightarrow \alpha + 3\beta = 1$

② $-\alpha - 5\beta = 3$

③ $\alpha + \beta = -1$

En ① $\alpha = 1 - 3\beta$

En ③ $1 - 3\beta + \beta = -1$

$-2\beta = -2 \rightarrow \beta = 1$

$\alpha = 1 - 3 \rightarrow \alpha = -2$

En ② $-\alpha - 5\beta = 3$

$-(-2) - 5(1) = 2 - 5 = -3$

Por tanto $p(x)$ NO es c.l. de $p_1(x)$ y $p_2(x)$

$p(x) = a_0 + a_1x + a_2x^2$ se puede escribir
como vector 3 componentes.

$$\begin{pmatrix} a_0 \\ a_1 \\ a_2 \end{pmatrix} \begin{matrix} \rightsquigarrow \text{ind} \\ \rightsquigarrow x \\ \rightsquigarrow x^2 \end{matrix} \quad \begin{pmatrix} a_2 \\ a_1 \\ a_0 \end{pmatrix} \begin{matrix} \rightsquigarrow x^2 \\ \rightsquigarrow x \\ \rightsquigarrow \text{ind} \end{matrix}$$

$$\begin{cases} 2\alpha + 6\beta = 2 \\ -\alpha - 5\beta = 3 \\ \alpha + \beta = -1 \end{cases}$$

$$\begin{pmatrix} 2 & 6 \\ -1 & -5 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} \alpha \\ \beta \end{pmatrix} = \begin{pmatrix} 2 \\ 3 \\ -1 \end{pmatrix}$$

$$\left(\begin{array}{cc|c} 2 & 6 & 2 \\ -1 & -5 & 3 \\ 1 & 1 & -1 \end{array} \right) \xrightarrow[\substack{f_2 + f_1 \\ f_3 - f_1}]{\substack{1/2 f_1}} \left(\begin{array}{cc|c} 1 & 3 & 1 \\ 0 & -2 & 4 \\ 0 & -2 & -2 \end{array} \right)$$

$$\xrightarrow{f_3 - f_2} \left(\begin{array}{cc|c} 1 & 3 & 1 \\ 0 & -2 & 4 \\ 0 & 0 & -6 \end{array} \right)$$

inconsistencia $0 = -6 ??$

$\beta = 1$
 $\alpha = -2$

$\alpha = -3$
e.l. de $p_1(x)$ y $p_2(x)$

Para hallar todas las o.l. posibles de $\mathbb{R}^2$

$$\hat{i} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad \hat{j} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

Un vector cualquiera $\begin{pmatrix} a \\ b \end{pmatrix} \in \mathbb{R}^2$

$$\begin{pmatrix} a \\ b \end{pmatrix} = \alpha \begin{pmatrix} 1 \\ 0 \end{pmatrix} + \beta \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

$$\begin{pmatrix} a \\ b \end{pmatrix} = a\hat{i} + b\hat{j}$$

hallar todas las o.l. posibles

a) $\begin{pmatrix} 1 \\ 2 \end{pmatrix}, \begin{pmatrix} -1 \\ 3 \end{pmatrix}$ b) $\begin{pmatrix} -1 \\ 2 \end{pmatrix}$ y $\begin{pmatrix} 2 \\ -4 \end{pmatrix}$

Soln:

a) $\begin{pmatrix} a \\ b \end{pmatrix} = \alpha \begin{pmatrix} 1 \\ 2 \end{pmatrix} + \beta \begin{pmatrix} -1 \\ 3 \end{pmatrix}$

$$\begin{pmatrix} 1 & -1 & : & a \\ 2 & 3 & : & b \end{pmatrix} \xrightarrow{[2]-2[1]} \begin{pmatrix} 1 & -1 & : & a \\ 0 & 5 & : & b-2a \end{pmatrix}$$

$$5\beta = b - 2a$$

$$\alpha - \beta =$$

$$\alpha = a +$$

$$\alpha = 3a$$

$$\begin{pmatrix} a \\ b \end{pmatrix} = \frac{3a +}{5}$$

$$5\beta = b - 2a \rightarrow \beta = \frac{b - 2a}{5}$$

$$\alpha - \beta = a$$

$$\alpha = a + \beta = a + \frac{b - 2a}{5}$$

$$\alpha = \frac{5a + b - 2a}{5}$$

$$\alpha = \frac{3a + b}{5}$$

$$\begin{pmatrix} a \\ b \end{pmatrix} = \frac{3a + b}{5} \begin{pmatrix} 1 \\ 2 \end{pmatrix} + \frac{b - 2a}{5} \begin{pmatrix} -1 \\ 3 \end{pmatrix}$$

Prueba: $\begin{pmatrix} 2 \\ -1 \end{pmatrix} \quad a = 2 \quad b = -1$

$$\alpha = \frac{3 \cdot 2 - 1}{5} \Rightarrow \boxed{\alpha = 1}$$

$$\beta = \frac{-1 - 2 \cdot 2}{5} = \frac{-1 - 4}{5} \Rightarrow \boxed{-1 = \beta}$$

$$\begin{pmatrix} 2 \\ -1 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \end{pmatrix} - \begin{pmatrix} -1 \\ 3 \end{pmatrix}$$

$$b) \begin{pmatrix} -1 & 2 & a \\ 2 & -4 & b \end{pmatrix} \xrightarrow{2+2f_1} \begin{pmatrix} -1 & 2 & a \\ 0 & 0 & b+2a \end{pmatrix}$$

Para que no se de una inconsistencia

$$b + 2a = 0$$

$$b = -2a$$

$$\begin{pmatrix} 2 \\ 6 \end{pmatrix} \text{ no es a.l. } 6 \neq -4$$

En (a) los 2 vectores me generan todo el espacio

$$\text{gen}\{v_1, v_2\} = \mathbb{R}^2$$

en (b) los vectores v_1, v_2 no generan todo el espacio, ellos generan un subespacio el que cumple la condición $b + 2a = 0$

$$H = \left\{ \begin{pmatrix} a \\ b \end{pmatrix} \in \mathbb{R}^2 \mid \underline{b + 2a = 0} \right\}$$

Restricción del espacio

$$\begin{pmatrix} a \\ -2a \end{pmatrix}$$

$$5\beta = b - 2a \rightarrow \beta = \frac{b}{5}$$

$$\alpha - \beta = a$$

$$\alpha = a + \beta = \underline{a + \frac{b}{5}}$$

$$\alpha = \underline{\frac{5a + b}{5}}$$

$$\alpha = \frac{3a + b}{5}$$

$$\begin{pmatrix} a \\ b \end{pmatrix} = \frac{3a + b}{5} \begin{pmatrix} 1 \\ 2 \end{pmatrix} +$$